Autodesk® Scaleform®

Scaleform 3.3   ,   ,   .

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:   4.01

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Autodesk® GAMEWARE

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Autodesk® Scaleform® 4.4

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Autodesk® Scaleform® 3.0 (heap) ,
. Scaleform 3.2
AMP(Analyzer for Memory and Performance)
. Scaleform 3.3
. 1.2
.

Scaleform 3.3 ,
.

-
- Movies Scaleform
- ActionScript (Garbage Collection)
-

1.1 중요한 메모리 개념

Scaleform ,
. / Scaleform
.

1.1.1 오버라이드 할당자

Scaleform GFx::System
.
.

1. Scaleform::SysAlloc Alloc/Free/Realloc . dlmalloc .

2. Scaleform , . " " .

3. Scaleform (: Scaleform::SysAllocWinAPI) . . (2) . 2 ' ' .

1.1.2 메모리 힙

, Scaleform , AMP MemReport . .

- Global - .
- MovieData - SWF/GFx .
- MovieView - GFx::Movie ActionScript . .
- MeshCashe - shape triangle data .

" " , . .

- .
- .
- .

- .

, .

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" "

.

Scaleform 3.3 Scaleform::SysAlloc

. , Scaleform 3.3 GFx::Movie

,

.

1.1.3 조각모음

Scaleform 3.0 AS(ActionScript) AS

. CLIK ActionScript

.

Scaleform GFx::Movie , AS

. Scaleform 3.3 Movie ,

(garbage collector) . ActionScript

Movie ,

Scaleform .

4: .

1.1.4 메모리 보고 및 디버깅

Scaleform "stat ID" ,

. stat ID stat ID

. Scaleform 3.2

AMP(Analyzer for Memory and Performance)

. AMP AMP .

, Scaleform::MemoryHeap::MemReport

.

stat ID . ,

. Scaleform

Visual Studio . Scaleform

, Scaleform

. Shipping

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1.2 Scaleform 3.3 메모리 API 변경점

Scaleform 3.3

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- GFx::Movie movie view

  .

  . 4 .

- SysAlloc "malloc " , 4K

  512

  SysAlloc .

  .

Scaleform::SysAlloc Scaleform::SysAllocPaged

. SysAlloc

SysAllocPaged . SysAllocStatic

, .

2 메모리 할당

1.1.1 , Scaleform

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1. SysAlloc
2. Scaleform
3. Scaleform

” ’. E E ῳ ǂ

SWF y K

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2.1 할당 오버라이딩

Scaleform .

1) SysAlloc . Alloc, Free, Realloc

2) Scaleform GFx::System

malloc/free Scaleform::SysAllocMalloc Scaleform .

2.1.1 자신만의 SysAlloc 구현

Scaleform SysAllocMalloc . SysAllocMalloc .

```
class MySysAlloc : public SysAlloc
{
  public:
    virtual void* Alloc(UPInt size, UPInt align)
    {
        return _aligned_malloc(size, align);
    }

    virtual void  Free(void* ptr, UPInt size, UPInt align)
    {
        SF_UNUSED2(size, align);
        _aligned_free(ptr);
        return true;
    }
    virtual void* Realloc(void* oldPtr, UPInt oldSize,
                          UPInt newSize, UPInt align)
    {
        SF_UNUSED(oldSize);
        return _aligned_realloc(oldPtr, newSize, align);
    }
 };
```

. MySysAlloc SysAlloc

3 (`Alloc, Free, Realloc`).

, Scaleform 16 . oldSize align

Free/Realloc . , Alloc/Free

wrapper Realloc .

, Scaleform

GFx::System .

```
MySysAlloc myAlloc;
GFxSystem gfxSystem(&myAlloc);
```

Scaleform 3.0 GFx::System Scaleform ,

Scaleform . Scaleform

. , Scaleform

(GFx::System **않는**).

GFx::System::Init() GFx::System::Destroy()

. GFx::System GFx::System::Init()

SysAlloc .

2.2 Scaleform 에 고정 메모리 블록 사용하기

1

, SysAlloc Scaleform .

SysAllocStatic .

```
void*            pmemChunk = …;
SysAllocStatic   blockAlloc(pmemChunk, 6*1024*1024);
System           gfxSystem(&blockAlloc);
…
```

6 Scaleform . ,

gfxSystem blockAlloc .

, " "

. ,

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Scaleform

.

. SysAllocStatic Alloc 0 , Scaleform

. SysAlloc Paged

.

2.2.1 메모리 아레나 (Memory Arenas)

Scaleform 3.1

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Scaleform GFx::Movie

Scaleform

. , , , Scaleform

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Scaleform

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Scaleform

. SDK

Scaleform . Scaleform 3.1

SWF/GFX .

Scaleform::Memory::DestroyArena

.

2.2.1.2 메모리 아레나의 사용

.

1. Scaleform::Memory::CreateArena 0(0 l

) .

   SysAlloc SysAllocStatic

   .

```
// Assume pbuf1 points to a buffer of 10,000,000 bytes.
 SysAllocStatic  sysAlloc(pbuf1, 10000000);
 Memory::CreateArena(1, &sysAlloc);
```

2. ID GFx::Loader::CreateMovie/GFx::MovieDef::CreateInstance

   . movie

   . “ ”

   .

```
// use arena 1 for the movie
Ptr<MovieDef> pMovieDef =
```

```
            *Loader.CreateMovie(filenameStr, Loader::LoadWaitFrame1, 1);
...
// use arena 1 for the instance
Ptr<Movie> pMovie =
        *pnewMovieDef->CreateInstance(MovieDef::MemoryParams(1), false);
```

3. GFx::MovieDef/GFx::Movie .

4. . GFx::MovieDef::WaitForLoadFinish

. GFx::ThreadedTaskManager

,

.

```
pMovieDef->WaitForLoadFinish(true);
pMovie    = 0;
pMovieDef = 0;
```

(Scaleform::Ptr):

```
pMovie->Release();
pMovieDef->Release();
```

5. DestroyArena . , CreateArena

. Scaleform::Memory::ArenaIsEmpty

, .

```
// DestroyArena will ASSERT if memory was not properly released before
// the call. In Release it will crash at "return *(int*)0;".
Memory::DestroyArena(1);
```

SysAllocobject 'pbuff1' (scope)

. .

2.3 *OS*

Scaleform OS

. .

- *Scaleform::SysAllocWinAPI* –VirtualAlloc()/VirtualFree() API . MS

  .

- *Scaleform::SysAllocPS3* –PS3 sys_memory_allocate/sys_memory_free

.

. OS

.

,

.

Scaleform WinAPI PS3 SysAlloc 64K

. , dlmalloc

2MB .

2.3.1 SysAllocPaged 구현하기

Scaleform 3.3 SysAlloc , SysAlloc

SysAllocPaged . SysAllocPaged

Scaleform , SysAllocStatic OS .

, . ,

SysAlloc SysAllocPaged .

SysAllocPagedMalloc .

```
class MySysAlloc : public SysAllocPaged
{
  public:
    virtual void  GetInfo(Info* i) const
    {
        i->MinAlign     = 1;
        i->MaxAlign     = 1;
        i->Granularity  = 128*1024;
        i->HasRealloc   = false;
    }

    virtual void* Alloc(UPInt size, UPInt align)
    {
        // Ignore 'align' since reported MaxAlign is 1.
        return malloc(size);
```

```
    }

    virtual bool  Free(void* ptr, UPInt size, UPInt align)
    {
        // free() doesn't need size or alignment of the memory block, but
        // you can use it in your implementation if it makes things easier.
        free(ptr);
        return true;
    }
};
```

. MySysAlloc SysAllocPaged

3 (`GetInfo, Alloc, Free`). ReallocInPlace

, .

- `GetInfo()` –Scaleform::SysAllocPaged::Info

  .

- `Alloc` – .

  . GetInfo MaxAlign . MaxAlign 1

  .

- `Free` –Alloc . free

  . .

- `ReallocInPlace` – .

  false . .

  [Scaleform Reference Documentation](#) .

`Alloc` `Free` .

`GetInfo` . Scaleform .

SysAllocPaged . , Scaleform

. `SysAllocPaged::Info` 4 .

- `MinAlign` –

- `MaxAlign` – . 0 Scaleform
  . 1 Alloc
  .
- `Granularity` –Scaleform . Scaleform
  . (malloc) 64K
- `HasRealloc` –`ReallocInPlace` . false
- `SysDirectThreshold` - null .
  SysDirectThreshold ,
  .
- `MaxHeapGranularity` - null , MaxHeapGranularity
  . , alloc/free
  MaxHeapGranularity .
  MaxHeapGranularity 4096 , 4096 .

`MaxAlign` SysAlloc .
. 3
.

1. .
   `MaxAlign` 1 Scaleform .
2. . `MaxAlign` 0
   . `Alloc` .
3. , .
   `MinAlign` `MaxAlign`
   OS .

Scaleform GFx::System

.

```
MySysAlloc myAlloc;
GFx::System gfxSystem(&myAlloc);
```

2.4

Scaleform . mesh

cache

. Scaleform API

.

2.4.1 힙 API

.

Scaleform ‘new’ .

```
Ptr<FileOpener> opener = *new FileOpener();
loader.SetFileOpener(opener);
```

‘new’ , Scaleform

. . (global)

GFx::System , 0

.

. ,

GFx::MeshCacheManager shape

. `GFx::Loader::CreateMovie()` SWF/GFX

. `GFx::MovieDef::CreateInstance`

. (scene) Scaleform

, .

. Scaleform

.

Scaleform

Scaleform::MemoryHeap . `Scaleform::MemoryHeap::CreateHeap`

`Scaleform::MemoryHeap::Release`

. `Scaleform::MemoryHeap::Alloc`

`Scaleform::MemoryHeap::Free` .

. SysAlloc

.

Scaleform

Scaleform::MemoryHeap .

- SF_HEAP_ALLOC(heap, size, id)
- SF_HEAP_MEMALIGN(heap, size, alignment, id)
- SF_HEAP_NEW(heap) ClassName
- SF_FREE(ptr)

.

'id' Scaleform AMP .

GFx::SpriteDef pHeap

.

```
Ptr<SpriteDef> def = *SF_HEAP_NEW(pHeap) SpriteDef(pdataDef);
```

GFx::SpriteDef Scaleform . Scaleform

Ptr<>

. Scaleform::NewOverrideBase

'delete' . delete SF_FREE(address)

.

2.4.2 자동 힙

,

. Scaleform::String/Scaleform::Array<>

. ,

.

```
class Sprite : public RefCountBase<Sprite>
{
    String                  Name;
    Array<Ptr<Sprite> > Children;

    Sprite(String& name, …) { … }
};
Ptr<Sprite> sprite = *SF_HEAP_NEW(pHeap) Sprite(name);
sprite->DoSomething();
```

GFx::Sprite ,

. , .

GFx::Sprite ,

. , ,

,

. Scaleform " " .

- SF_HEAP_AUTO_ALLOC(ptr, size)
- SF_HEAP_AUTO_NEW(ptr) ClassName

SF_HEAP_ALLOC SF_HEAP_NEW . ,

. ,

, . .

```
MemoryHeap*    pheap = …;
UByte *          pbuffer = (UByte*)SF_HEAP_ALLOC(pheap, 100,
StatMV_ActionScript_Mem);
Ptr<Sprite> sprite = *SF_HEAP_AUTO_NEW(pbuffer + 5) Sprite();
…
sprite->DoSomething();
...
// Release sprite and free buffer memory.
sprite = 0;
SF_FREE(pbuffer);
```

, GFx::Sprite .

'pbuffer' SF_HEAP_AUTO_NEW ,

. , Scaleform

.

.

.

,

. , GFx::Sprite

.

```
class Sprite : public RefCountBase<Sprite>
{
    StringLH                    Name;
    ArrayLH<Ptr<Sprite> > Children;

    Sprite(String& name, …) { … }
};
Ptr<Sprite> sprite = *SF_HEAP_NEW(pHeap) Sprite(name);
```

```
sprite->DoSomething();
```

Scaleform::String Scaleform::Array<> " (Local Heap)"

Scaleform::StringLH Scaleform::ArrayLH<> .

" " .

. Scaleform

.

2.4.3 힙 추적기(Heap Tracer)

GFxConfig.h SF_MEMORY_TRACE_ALL

Scaleform::Memory SetTracer CreateHeap, DestroyHeap, Alloc, Free

.

SysAllocWinAPI .

```
#include "Kernel/HeapPT/HeapPT_SysAllocWinAPI.h"

class MyTracer : public Scaleform::MemoryHeap::HeapTracer
{
 public:
    virtual void OnCreateHeap(const MemoryHeap* heap)
    {
        //...
    }

    virtual void OnDestroyHeap(const MemoryHeap* heap)
    {
        //...
    }

    virtual void OnAlloc(const MemoryHeap* heap, UPInt size, UPInt align,
                           unsigned sid, const void* ptr)
    {
        //...
    }

    virtual void OnRealloc(const MemoryHeap* heap, const void* oldPtr,
                          UPInt  newSize, const void* newPtr)
    {
        //...
    }
```

```
    virtual void OnFree(const MemoryHeap* heap, const void* ptr)
    {
        //...
    }
};

static MyTracer tracer;
static SysAllocWinAPI sysAlloc;
static GFx::System* gfxSystem;


class MySystem : public Scaleform::System
{
 public:

    MySystem() : Scaleform::System(&sysAlloc)
    {
        Scaleform::Memory::GetGlobalHeap()->SetTracer(&tracer);
    }
    ~MySystem() { }
};


SF_PLATFORM_SYSTEM_APP(FxPlayer, MySystem, FxPlayerApp)
```

3 메모리 보고

Scaleform Player AMP HUD Scaleform 3.0 ,

PC Scaleform

AMP . AMP AMP

.

Scaleform Player HUD UI , (Ctrl + F5)

. Scaleform::MemoryHeap::MemReport

.

```
void MemReport (class StringBuffer& buffer, MemReportType detailed,
                    bool xmlFormat = false);
void MemReport (struct MemItem* rootItem, MemReportType detailed);
```

MemReport , XML

. .

MemReportType .

- MemoryHeap::MemReportBrief – Scaleform .

  .

```
Memory 4,680K / 4,737K
    Image                                     1,052
    Sound                                     136
    Movie View                                1,739,784
    Movie Data                                300,156
```

- MemoryHeap::MemReportFull – Ctrl+F6 Scaleform Player

  .

  SF_MEMORY_ENABLE_DEBUG_INFO stat ID memory

  breakdown . .

```
Memory 4,651K / 4,707K
    System Summary
        System Memory FootPrint                     4,832,440
        System Memory Used Space                    4,775,684
        Debug Heaps Footprint                       12,884
        Debug Heaps Used Space                      5,392
    Summary
        Image                                       1,052
        Sound                                       136
        Movie View                                  1,715,128
        Movie Data                                  300,156
    [Heap] Global                                   4,848,864
        Heap Summary
            Total Footprint                         4,848,864
            Local Footprint                         2,427,860
            Child Footprint                         2,421,004
            Child Heaps                             4
            Local Used Space                        2,417,196
            DebugInfo                               120,832
        Memory
            MovieDef
                Sounds                              8
                ActionOps                           80,476
                MD_Other                            200
            MovieDef                                36
            MovieView
                MV_Other                            32
            General                                 91,486
            Image                                   92
            Sound                                   136
            String                                  31,425
            Debug Memory
                StatBag                             16,384
            Renderer
                Buffers                             2,097,152
                RenderBatch                         5,696
                Primitive                           1,512
                Fill                                900
                Mesh                                4,884
                MeshBatch                           2,200
                Context                             15,184
                NodeData                            15,160
                TreeCache                           13,220
                TextureManager                      2,336
                MatrixPool                          4,096
                MatrixPoolHandles                   2,032
                Text                                1,232
```

```
        Renderer                                  13,488
    Allocations Count                             1,822
```

- MemoryHeap:: MemReportFileSummary –

. SWF MovieData, MovieDef MovieViews stat ID

. .

```
Movie File 3DGenerator_AS3.swf
    Memory                                          1,956,832
        MovieDef                                    152,664
            CharDefs                                39,848
            ShapeData                               1,716
            Tags                                    73,656
            MD_Other                                37,444
        MovieView                                   196,249
            MovieClip                               120
            ActionScript                            181,917
            MV_Other                                14,212
        General                                     1,596,712
        Image                                       960
        String                                      2,783
        Renderer                                    7,464
            Text                                    7,464
```

- MemoryHeap::MemReportHeapDetailed – AMP

, ,

Scaleform . :

```
Total Footprint                                 4,858,148
    Used Space                                  4,793,972
        Global Heap                             2,405,492
            MovieDef                            80,720
                Sounds                          8
                ActionOps                       80,476
                MD_Other                        200
            MovieView                           32
                MV_Other                        32
            General                             92,566
            Image                               92
            Sound                               136
            String                              31,276
            Debug Memory                        8,192
                StatBag                         8,192
            Renderer                            2,182,960
```

```
            Buffers                        2,097,152
            RenderBatch                    5,696
            Primitive                      1,512
            Fill                           900
            Mesh                           4,884
            MeshBatch                      2,200
            Context                        15,184
            NodeData                       15,016
            TreeCache                      13,136
            TextureManager                 2,336
            MatrixPool                     8,192
            MatrixPoolHandles              2,032
            Text                           1,232
    Movie Data Heaps                       300,156
        MovieData "3DGenerator_AS3.swf"    300,156
            MovieDef                       152,664
                CharDefs                   39,848
                ShapeData                  1,716
                Tags                       73,656
                MD_Other                   37,444
            General                        141,332
            Image                          960
            String                         2,312
    Movie View Heaps                       1,727,936
        MovieView "3DGenerator_AS3.swf"    1,727,936
            MovieView                      195,977
                MovieClip                  120
                ActionScript               181,645
                MV_Other                   14,212
            General                        1,467,668
            String                         471
            Renderer                       7,464
                Text                       7,464
    Other Heaps                            360,580
        _FMOD_Heap                         360,580
            General                        359,944
Debug Data                                 12,884
Unused Space                               51,292
    Global                                 51,292
        _FMOD_Heap                         4,384
        MovieData "3DGenerator_AS3.swf"    5,356
        MovieView "3DGenerator_AS3.swf"    33,032
```

Scaleform SF_MEMORY_ENABLE_DEBUG_INFO

"stat ID" .

- MovieDef – .
  .
- MovieView – GFx::Movie . ActionScript
  movie clip
  .
- CharDefs – movie clip flash .
- ShapeData – .
- Tags – keyframe .
- ActionOps – ActionScript .
- MeshCache – tesselator EdgeAA
  . .
- FontCache – .

4 가베지 콜렉션

Scaleform 2.0 .

. , 2 (circular

references) . .

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```
Code:
var o1 = new Object;
var o2 = new Object;
o1.a = o2;
o2.a = o1;
```

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Scaleform 3.0

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Scaleform 3.0

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. , Scaleform

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Include/GFxConfig.h

GFX_AS_ENABLE_GC .

```
// Enable garbage collection
#define GFX_AS_ENABLE_GC
```

Scaleform .

4.1

Scaleform 3.0 . Scaleform 3.3

, MovieView .

MovieDef .

```
Movie* MovieDef::CreateInstance(const MemoryParams& memParams,
                                                   bool initFirstFrame = true)
```

MemoryParams .

```
struct MemoryParams
{
      MemoryHeap::HeapDesc    Desc;
      float                   HeapLimitMultiplier;
      unsigned                MaxCollectionRoots;
      unsigned                FramesBetweenCollections;
};
```

- Desc – . (MinAlign), (Granularity), (Limit), (Flags) . (Limit) , Scaleform .
- HeapLimitMultiplier – (0~1). . .
- MaxCollectionRoots – roots , root . . .
- FramesBetweenCollections – roots . root , . .

movie view .

```
MemoryContext* MovieDef::CreateMemoryContext(
                                              const char* heapName,
                                              const MemoryParams& memParams,
```

```
                                             bool debugHeap)

Movie* MovieDef::CreateInstance(MemoryContext* memContext,
                                            bool initFirstFrame = true)
```

MemoryContext movie view

. movie view movie view ,

- (thread-safe) AMP

AMP .

movie view , ,

.

`MemoryParams::Desc`

. (:ActionScript) (`Desc.Limit`, `HeapLimitMultiplier`)

. 128K .(

.) , .

, . Boehm-Demers-Weiser(BDW)

-

. + Delta

( ).

.

`MaxCollectionRoots`/ `FramesBetweenCollections`. `MaxCollectionRoots`

root . “root”

.

( ) root . (: “obj.member=1” “obj” .)

MaxCollectionRoots .

0 .

`FramesBetweenCollections` .

“frame” .

.

`FramesBetweenCollections` 1800 30fps

60 . `DescLimit` ,

`FramesBetweenCollections` .

0 , .

4.2 GFx::Movie:: ForceCollectGarbage

```
virtual void  ForceCollectGarbage() = 0;
```

.

.